

# Sandeep Ganesan

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## EDUCATION

**Ritsumeikan University (立命館大学)**

*B.Eng. Information Systems Science and Engineering, GPA: 4.5/5.0*

Ibaraki-shi, Osaka

*Apr. 2024 – Mar 2028*

## TECHNICAL SKILLS

**Languages:** Python, C/C++, Dart, JavaScript/TypeScript, Java, HTML/CSS

**Frameworks:** ROS 2, React, React Native, Flutter, Astro, Flask

**Developer Tools:** Git, Firebase, Firestore, Docker, Cloudflare, Google Cloud Platform, Linux, CMake, LaTeX, Gazebo, CAD

**Libraries & AI:** PyTorch, TensorFlow, OpenCV, BERT, Retrieval-Augmented Generation (RAG), SLAM, Cartographer, Nanostores, pandas, NumPy

## PROJECTS

**Portfolio Website** | *Astro, React, Tailwind CSS, Nanostores*

May 2026 – Present

- Designed a fast, retro-themed portfolio shipping zero-JavaScript static HTML by default
- Built a custom React window manager featuring draggable bounds and focus hierarchy
- Achieved decoupled global state management across UI components using Nanostores
- Integrated complex React "Islands" while maintaining optimal site loading performance

**読READER (YomiREADER)** | *React Native, Llama 3.2, llama.rn, C++, Git*

Jan 2026 – Present

- Architected a completely offline EPUB reader with an on-device AI assistant via llama.rn
- Built an on-device RAG pipeline using Nomic Embed for context-aware book querying
- Accelerated local processing on mobile hardware by developing custom vector caching
- Enhanced the user experience with a custom reader core and gesture navigation

**Streamoo** | *Flutter, Flask, Firebase, Firestore, Git*

Jun 2025 – Present

- Built a cross-platform tracking app providing real-time visibility of Twitch and YouTube creators
- Engineered a polling-based Flask backend to query streaming APIs every 15 minutes
- Synced live updates directly to Firestore to minimize client-side network overhead
- Enabled low-latency state changes for the mobile interface and home screen widgets

**Dynamic VAA Updater** | *Python, BERT, BERTopic, YouTube API, Semantic Analysis*

Oct 2024 – Jan 2025

- Co-developed an ML pipeline to update Voting Advice Applications based on social media activity
- Automated real-time data ingestion pipelines using the YouTube API and custom scrapers
- Filtered raw text and reduced data noise prior to processing using BERTopic modeling
- Fine-tuned a BERT model to flag and quantify shifts in political orientation

## EXPERIENCE

**Hardware Team Specialist**

June 2024 – Present

*Ri-one @home Club (Robotics Club)*

*Ritsumeikan University, Ibaraki-shi, Osaka*

- Won 3rd Place in the Open Challenge at Japan Open 2026, competing with a custom service robot named "Sudo"
- Utilized CAD software to design, prototype, and manufacture structural components for the robot's chassis
- Initialized and configured core sensors and hardware components within a unified ROS2 environment
- Developed autonomous "follow-me" navigation and human-tracking software routines
- Programmed interaction logic for the "Receptionist" and "Carry My Luggage" competition challenges

## AWARDS AND HONOURS

**Monbukagakusho Honors Scholarship**

Apr 2026 – Present

**Microsoft Azure AZ-900 Certification**

Sep 2025

**Monbukagakusho Honors Scholarship**

Apr 2024 – Mar 2025

**TOEFL - 103. of 120**

Dec 2022